

Proximity Measures



Fabio Stella

Department of Informatics, Systems and Communication

University of Milano-Bicocca

fabio.stella@unimib.it

OUTLOOK

- **SIMILARITY – DISSIMILARITY**
- **EUCLIDEAN – MINKOWSKI – CITY BLOCK – MAHALANOBIS**
- **SIMPLE MATCHING COEFFICIENT – JACCARD – COSINE SIMILARITY**
- **CORRELATION – ENTROPY – MUTUAL INFORMATION**
- **SELECTING THE RIGHT PROXIMITY MEASURE**

- **SIMILARITY MEASURE**

- numerical measure of how alike two data objects are
- is higher when objects are more alike
- often falls in the range $[0,1]$

- **DISSIMILARITY MEASURE**

- numerical measure of how different two data objects are
- lower when objects are more alike
- minimum dissimilarity is often 0
- upper limit varies

- **PROXIMITY** refers to a similarity or dissimilarity

The following table shows the similarity and dissimilarity between two objects, x and y , with respect to a single, simple attribute.

Attribute Type	Dissimilarity	Similarity
Nominal	$d = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$	$s = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$
Ordinal	$d = x - y / (n - 1)$ (values mapped to integers 0 to $n-1$, where n is the number of values)	$s = 1 - d$
Interval or Ratio	$d = x - y $	$s = -d, s = \frac{1}{1+d}, s = e^{-d},$ $s = 1 - \frac{d - \min_d}{\max_d - \min_d}$

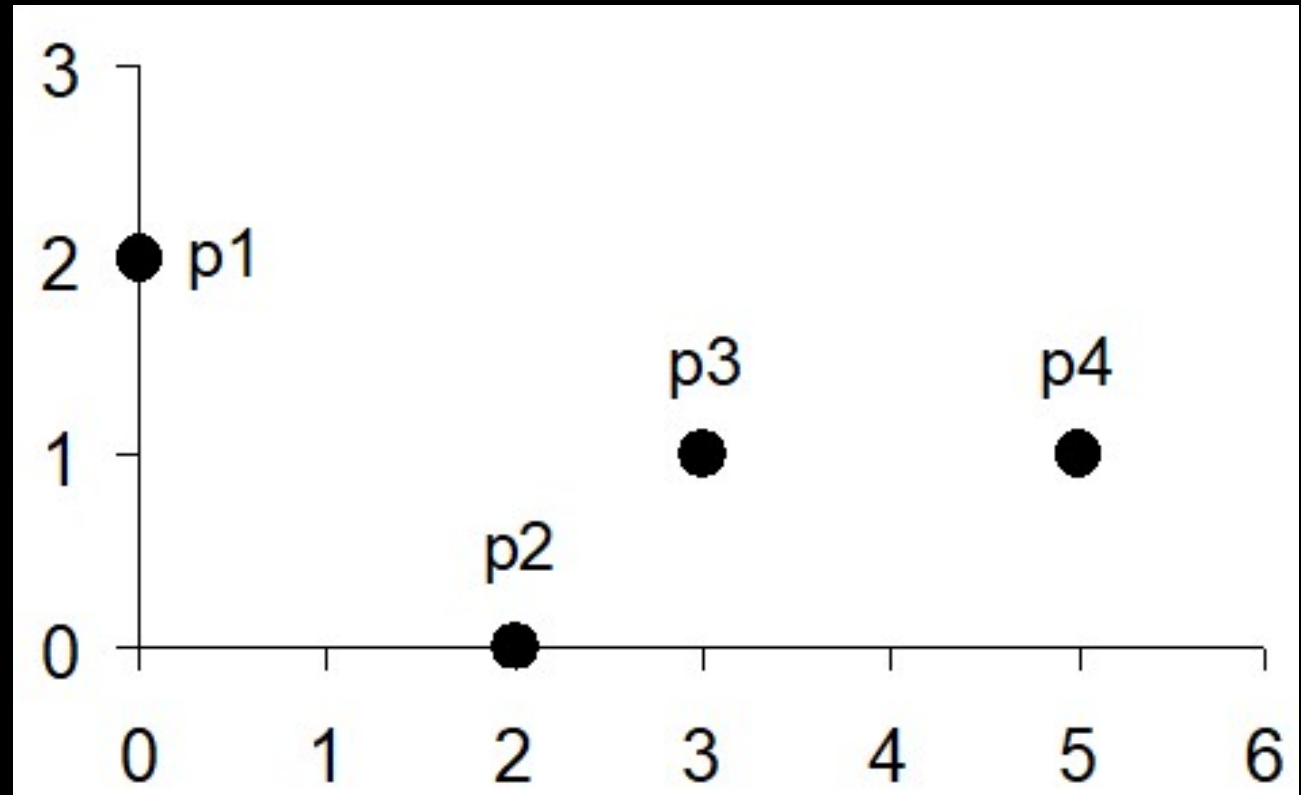
EUCLIDEAN DISTANCE

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}$$

where n is the number of dimensions (attributes) and x_k and y_k are, respectively, the k^{th} attributes (components) of data objects $\mathbf{x}$ and $\mathbf{y}$.

— standardization is necessary, if scales differ

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1



$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}$$

	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

Distance Matrix

MINKOWSKI DISTANCE is a generalization of euclidean distance

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{k=1}^n |x_k - y_k|^r \right)^{1/r}$$

where r is a parameter, n is the number of dimensions (attributes) and x_k and y_k are, respectively, the k^{th} attributes (components) of data objects $\mathbf{x}$ and $\mathbf{y}$.

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{k=1}^n |x_k - y_k|^r \right)^{1/r}$$

- $r = 1$. **CITY BLOCK** (**MANHATTAN**, **TAXICAB**, **L_1 norm**) distance

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{k=1}^n |x_k - y_k| \right)$$

A common example of this for binary vectors is the **HAMMING DISTANCE**, which is just the number of bits that are different between two binary vectors

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

L1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

(0, 0, 1, 0, 0, 1, 0, 1, 1, 0)

(1, 0, 1, 1, 0, 0, 0, 0, 1, 1)

$$1 + 0 + 0 + 1 + 0 + 1 + 0 + 1 + 0 + 1 = 5$$

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{k=1}^n |x_k - y_k|^r \right)^{1/r}$$

- $r = 2$. **EUCLIDEAN DISTANCE**

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}$$

- $r \rightarrow \infty$. **"SUPREMUM"** ($L_{\max}$ norm, L_{∞} norm) distance

This is the maximum difference between any component of the vectors

$$d(\mathbf{x}, \mathbf{y}) = \operatorname{argmax}_k |x_k - y_k|$$

Do not confuse r with n , i.e., all these distances are defined for all numbers of dimensions.

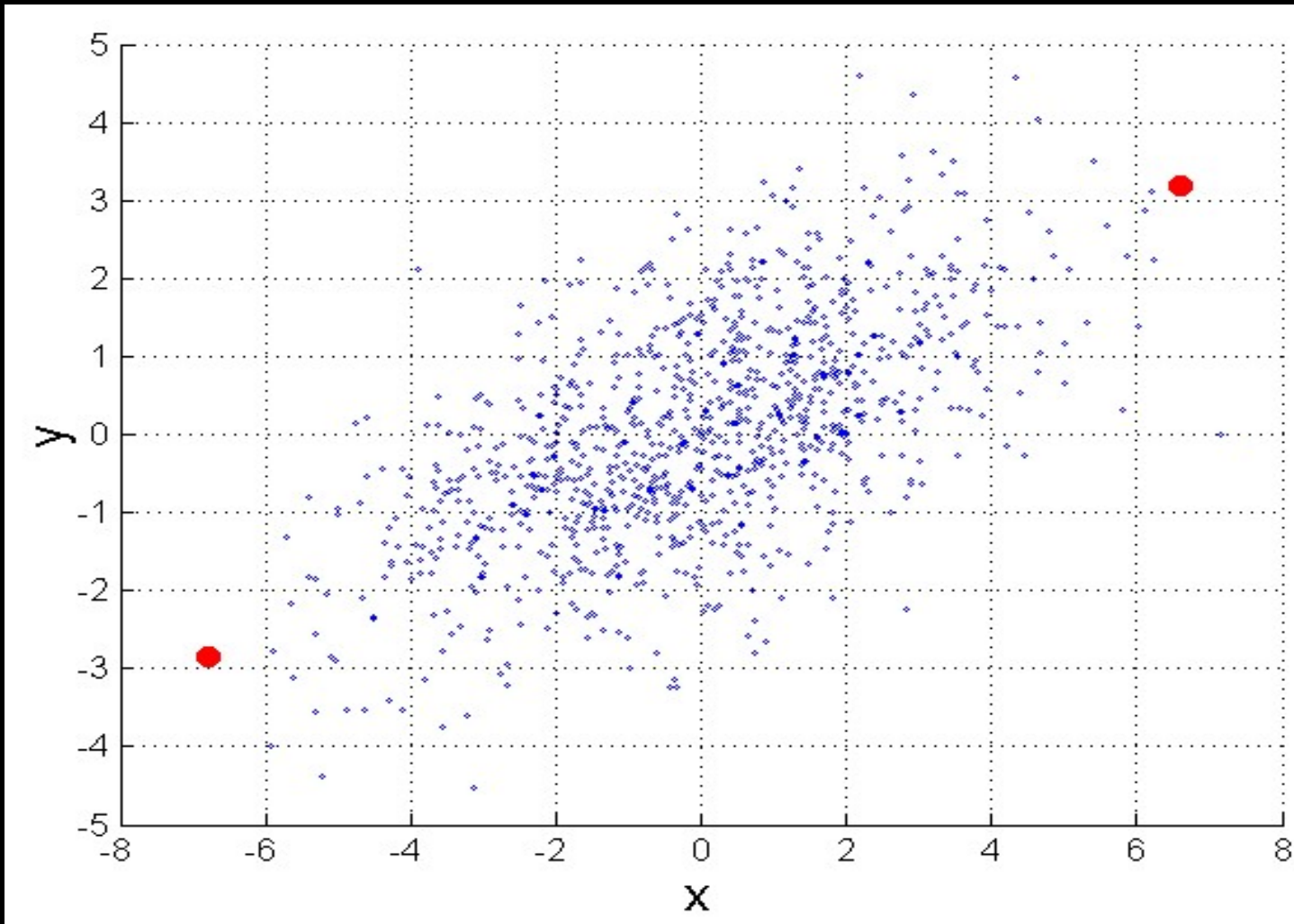
point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

L2	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

L_{∞}	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

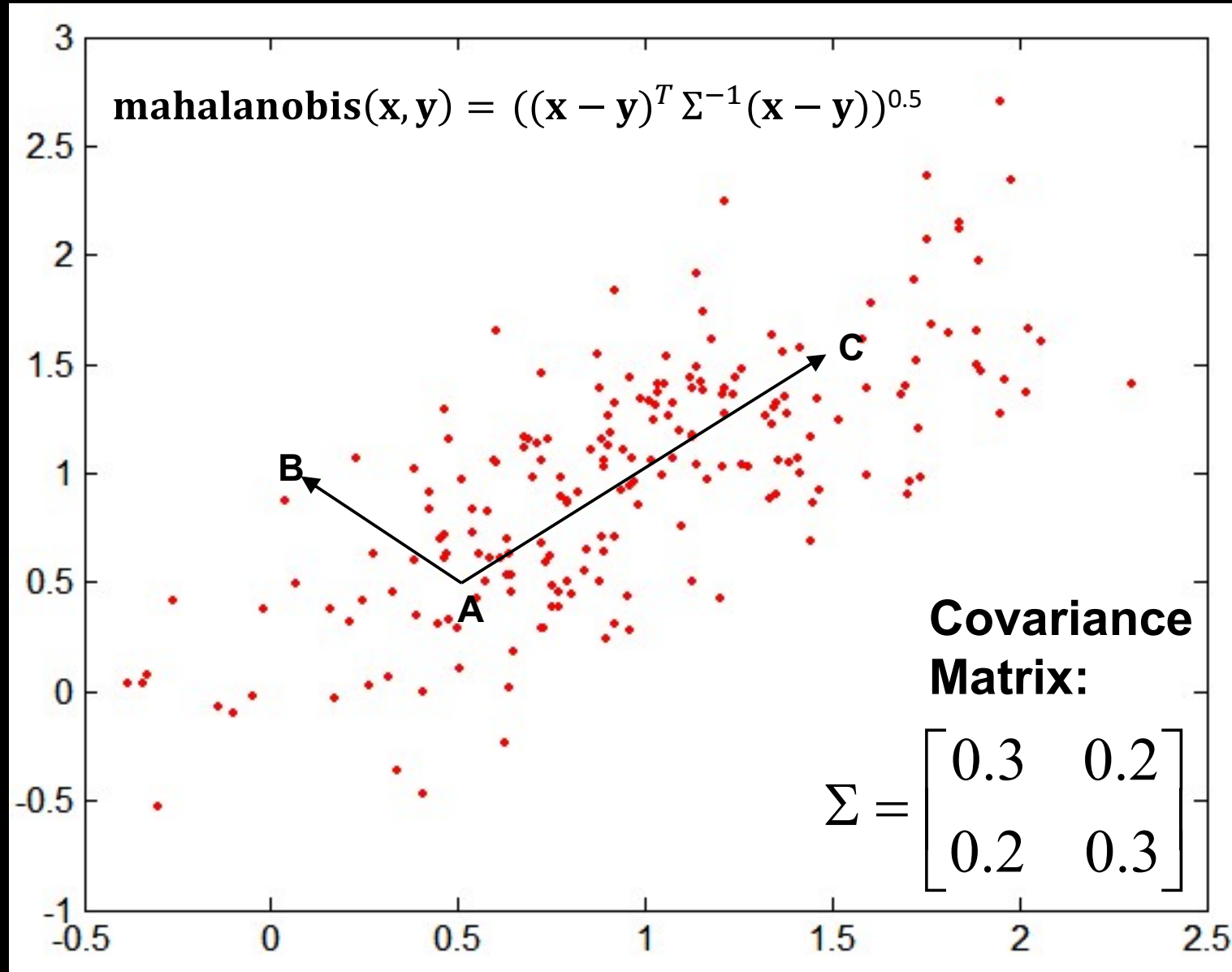
MAHALANOBIS DISTANCE

$$\text{mahalanobis}(\mathbf{x}, \mathbf{y}) = ((\mathbf{x} - \mathbf{y})^T \Sigma^{-1} (\mathbf{x} - \mathbf{y}))^{0.5}$$



Σ is the covariance matrix

For red points, the euclidean distance is 14.7, Mahalanobis distance is 6.



$$\mathbf{A} = (0.5, 0.5)$$

$$\mathbf{B} = (0, 1)$$

$$\mathbf{C} = (1.5, 1.5)$$

$$\text{Mahal}(\mathbf{A}, \mathbf{B}) = 5$$

$$\text{Mahal}(\mathbf{A}, \mathbf{C}) = 4$$

DISTANCES, such as the Euclidean distance, have some well known **PROPERTIES**.

- $d(\mathbf{x}, \mathbf{y}) \geq 0$ for all $\mathbf{x}$ and $\mathbf{y}$ and $d(\mathbf{x}, \mathbf{y}) = 0$ iff $\mathbf{x} = \mathbf{y}$ **NON NEGATIVE**
- $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ for all $\mathbf{x}$ and $\mathbf{y}$ **SYMMETRY**
- $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$ for all points $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ **TRIANGLE INEQUALITY**

where $d(\mathbf{x}, \mathbf{y})$ is the distance (dissimilarity) between points (data objects), $\mathbf{x}$ and $\mathbf{y}$

A distance that satisfies these properties is a **METRIC**

SIMILARITIES, also have some well known **PROPERTIES**.

— $s(\mathbf{x}, \mathbf{y}) = 1$ (or maximum similarity) only if $\mathbf{x} = \mathbf{y}$ (does not always hold, e.g., cosine)

— $s(\mathbf{x}, \mathbf{y}) = s(\mathbf{y}, \mathbf{x})$ for all $\mathbf{x}$ and $\mathbf{y}$ **SYMMETRY**

where $s(\mathbf{x}, \mathbf{y})$ is the similarity between points (data objects), $\mathbf{x}$ and $\mathbf{y}$.

- Common situation is that objects, $\mathbf{x}$ and $\mathbf{y}$, have only binary attributes
- Compute similarities using the following quantities
 - f_{01} = the number of attributes where $\mathbf{x}$ was 0 and $\mathbf{y}$ was 1
 - f_{10} = the number of attributes where $\mathbf{x}$ was 1 and $\mathbf{y}$ was 0
 - f_{00} = the number of attributes where $\mathbf{x}$ was 0 and $\mathbf{y}$ was 0
 - f_{11} = the number of attributes where $\mathbf{x}$ was 1 and $\mathbf{y}$ was 1

- **SIMPLE MATCHING COEFFICIENT**

$$\text{SMC} = \text{number of matches} / \text{number of attributes} = \frac{f_{11} + f_{00}}{f_{01} + f_{10} + f_{11} + f_{00}}$$

- **JACCARD COEFFICIENT**

$$J = \text{number of 11 matches} / \text{number of non-zero attributes} = \frac{f_{11}}{f_{01} + f_{10} + f_{11}}$$

$$\mathbf{x} = (1, 0, 0, 0, 0, 0, 0, 0, 0, 0)$$

$$\mathbf{y} = (0, 0, 0, 0, 0, 0, 1, 0, 0, 1)$$

$$f_{01} = 2 \quad (\text{the number of attributes where } \mathbf{x} \text{ was 0 and } \mathbf{y} \text{ was 1})$$

$$f_{10} = 1 \quad (\text{the number of attributes where } \mathbf{x} \text{ was 1 and } \mathbf{y} \text{ was 0})$$

$$f_{00} = 7 \quad (\text{the number of attributes where } \mathbf{x} \text{ was 0 and } \mathbf{y} \text{ was 0})$$

$$f_{11} = 0 \quad (\text{the number of attributes where } \mathbf{x} \text{ was 1 and } \mathbf{y} \text{ was 1})$$

$$\text{SMC} = \frac{f_{11} + f_{00}}{f_{01} + f_{10} + f_{11} + f_{00}} = \frac{0 + 7}{2 + 1 + 0 + 7} = 0.7$$

$$J = \frac{f_{11}}{f_{01} + f_{10} + f_{11}} = \frac{0}{2 + 1 + 0} = 0$$

If $\mathbf{d}_1$ and $\mathbf{d}_2$ are two document vectors, then

$$\cos(\mathbf{d}_1, \mathbf{d}_2) = \langle \mathbf{d}_1, \mathbf{d}_2 \rangle / \|\mathbf{d}_1\| \|\mathbf{d}_2\|, \quad \text{COSINE SIMILARITY}$$

where $\langle \mathbf{d}_1, \mathbf{d}_2 \rangle$ indicates inner product or vector dot product of vectors, $\mathbf{d}_1$ and $\mathbf{d}_2$, and $\|\mathbf{d}\|$ is the length of vector $\mathbf{d}$.

EXAMPLE: $\mathbf{d}_1 = (3, 2, 0, 5, 0, 0, 0, 2, 0, 0)$ $\mathbf{d}_2 = (1, 0, 0, 0, 0, 0, 0, 1, 0, 2)$

$$\langle \mathbf{d}_1, \mathbf{d}_2 \rangle = 3 \cdot 1 + 2 \cdot 0 + 0 \cdot 0 + 5 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 2 \cdot 1 + 0 \cdot 0 + 0 \cdot 2 = 5$$

$$\|\mathbf{d}_1\| = (3 \cdot 3 + 2 \cdot 2 + 0 \cdot 0 + 5 \cdot 5 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 2 \cdot 2 + 0 \cdot 0 + 0 \cdot 0)^{0.5} = (42)^{0.5} = 6.481$$

$$\|\mathbf{d}_2\| = (1 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 + 2 \cdot 2)^{0.5} = (6)^{0.5} = 2.449$$

$$\cos(\mathbf{d}_1, \mathbf{d}_2) = \langle \mathbf{d}_1, \mathbf{d}_2 \rangle / \|\mathbf{d}_1\| \|\mathbf{d}_2\| = 5 / (6.481 \cdot 2.449) = 0.3150$$

If $\mathbf{d}_1$ and $\mathbf{d}_2$ are two document vectors, then

$$\cos(\mathbf{d}_1, \mathbf{d}_2) = \langle \mathbf{d}_1, \mathbf{d}_2 \rangle / \|\mathbf{d}_1\| \|\mathbf{d}_2\|, \quad \text{COSINE SIMILARITY}$$

where $\langle \mathbf{d}_1, \mathbf{d}_2 \rangle$ indicates inner product or vector dot product of vectors, $\mathbf{d}_1$ and $\mathbf{d}_2$, and $\|\mathbf{d}\|$ is the length of vector $\mathbf{d}$.

EXAMPLE: $\mathbf{d}_1 = (3, 2, 0, 5, 0, 0, 0, 2, 0, 0)$ $\mathbf{d}_2 = (6, 4, 0, 10, 0, 0, 0, 4, 0, 0)$

$$\langle \mathbf{d}_1, \mathbf{d}_2 \rangle = 3 \cdot 6 + 2 \cdot 4 + 0 \cdot 0 + 5 \cdot 10 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 2 \cdot 4 + 0 \cdot 0 + 0 \cdot 0 = 84$$

$$\|\mathbf{d}_1\| = (3 \cdot 3 + 2 \cdot 2 + 0 \cdot 0 + 5 \cdot 5 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 2 \cdot 2 + 0 \cdot 0 + 0 \cdot 0)^{0.5} = (42)^{0.5} = 6.481$$

$$\|\mathbf{d}_2\| = (6 \cdot 6 + 4 \cdot 4 + 0 \cdot 0 + 10 \cdot 10 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 4 \cdot 4 + 0 \cdot 0 + 0 \cdot 0)^{0.5} = (166)^{0.5} = 12.88$$

$$\cos(\mathbf{d}_1, \mathbf{d}_2) = \langle \mathbf{d}_1, \mathbf{d}_2 \rangle / \|\mathbf{d}_1\| \|\mathbf{d}_2\| = 84 / (6.481 \cdot 12.88) = 1$$

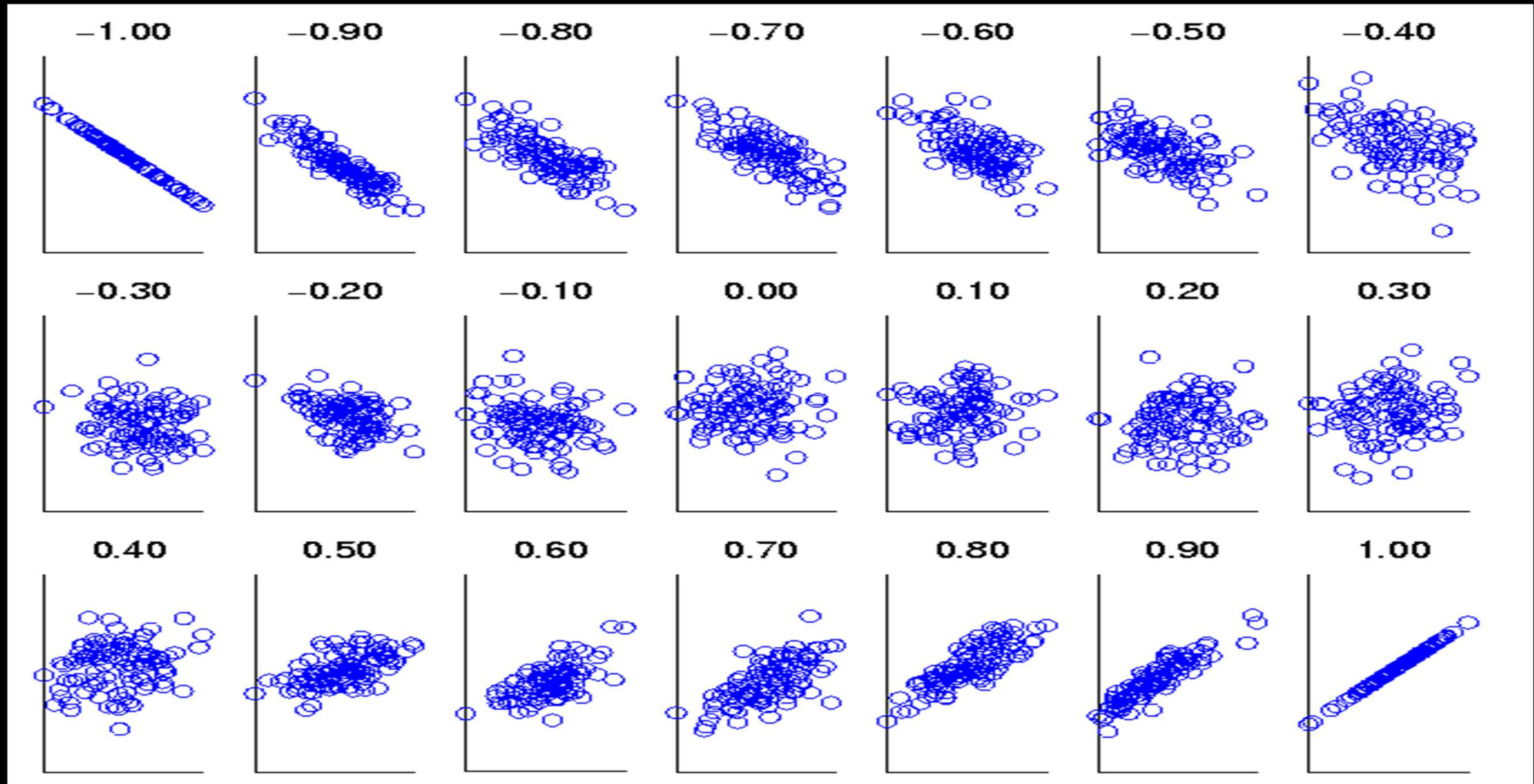
CORRELATION (PEARSON)

Measures the linear relationship between two sets of values that are observed together.

$$\text{Corr}(\mathbf{x}, \mathbf{y}) = \frac{\text{Cov}(\mathbf{x}, \mathbf{y})}{\sqrt{\text{Var}(\mathbf{x})\text{Var}(\mathbf{y})}} = \frac{s_{\mathbf{xy}}}{s_{\mathbf{x}} s_{\mathbf{y}}} \quad \mathbf{x}, \mathbf{y}, n\text{-dimensional vectors}$$

$$s_{\mathbf{xy}} = \frac{1}{n-1} \sum_{k=1}^n (\mathbf{x}_k - \bar{\mathbf{x}}) (\mathbf{y}_k - \bar{\mathbf{y}}) \quad s_{\mathbf{x}} = \sqrt{\frac{1}{n-1} \sum_{k=1}^n (\mathbf{x}_k - \bar{\mathbf{x}})^2} \quad s_{\mathbf{y}} = \sqrt{\frac{1}{n-1} \sum_{k=1}^n (\mathbf{y}_k - \bar{\mathbf{y}})^2}$$

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k \quad \bar{\mathbf{y}} = \frac{1}{n} \sum_{k=1}^n \mathbf{y}_k$$



DRAWBACK OF CORRELATION

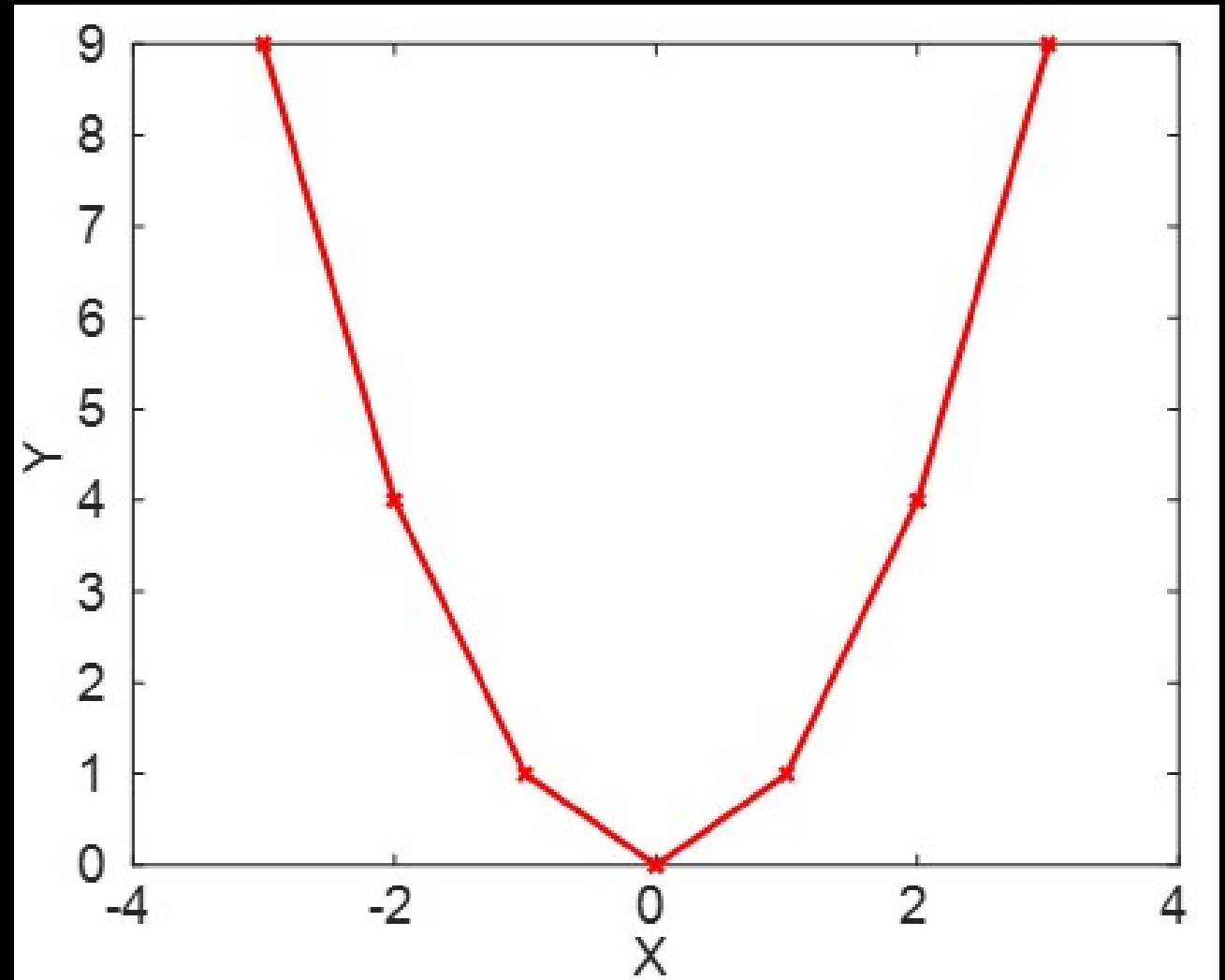
$$\mathbf{x} = (-3, -2, -1, 0, 1, 2, 3)$$

$$\mathbf{y} = (9, 4, 1, 0, 1, 4, 9)$$

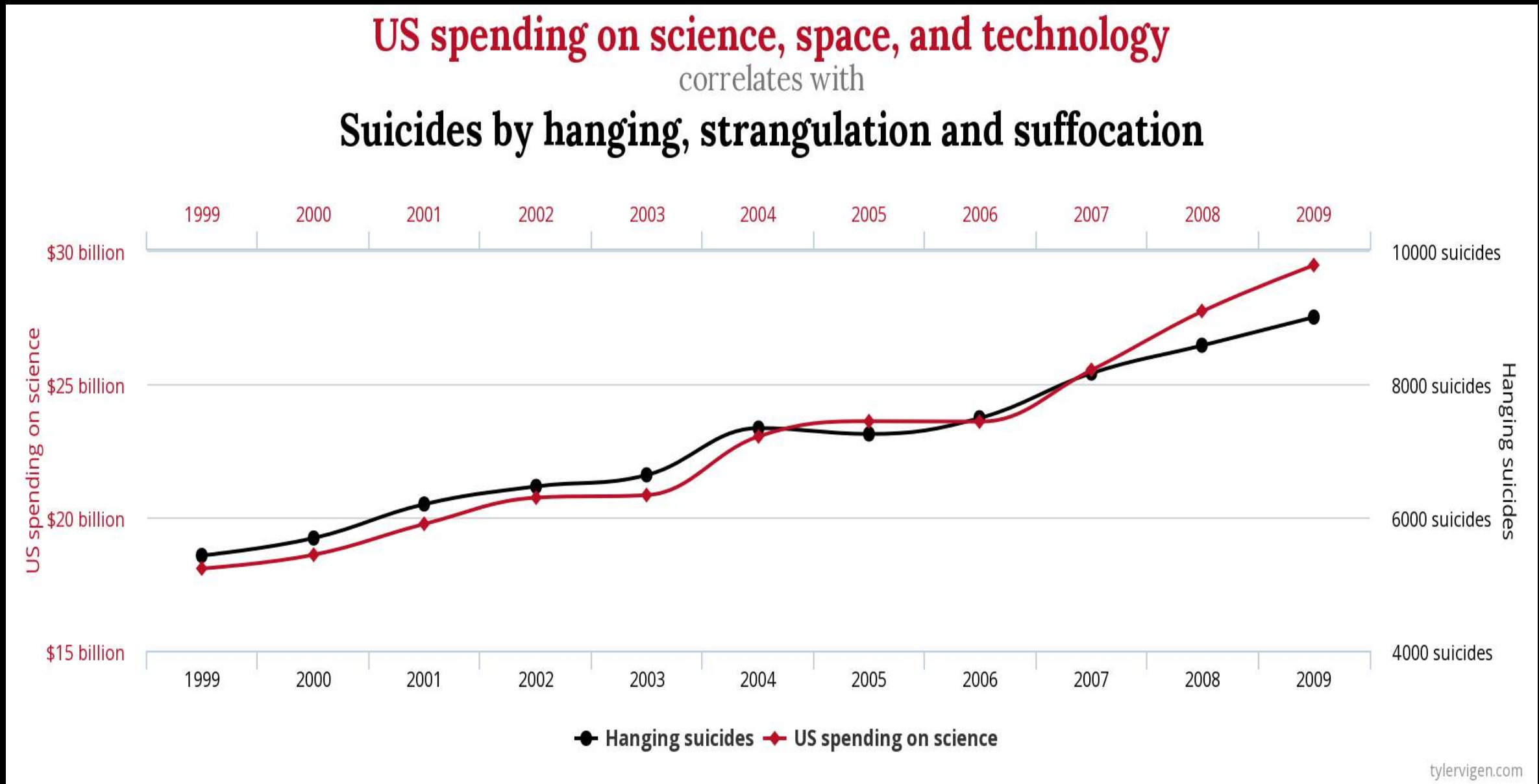
$$y_i = x_i^2$$

$$\bar{x} = 0, \bar{y} = 4$$

$$s_x = 2.16, s_y = 3.74$$



$$\text{Corr}(\mathbf{x}, \mathbf{y}) = (-3) \cdot (5) + (-2) \cdot (0) + (-1) \cdot (-3) + (0) \cdot (-4) + (1) \cdot (-3) + (2) \cdot (0) + (3) \cdot (5) / (6 \cdot 2.16 \cdot 3.74) = 0$$



Compare the three proximity measures according to their behavior under variable transformation

- **SCALING**: multiplication by a value
- **TRANSLATION**: adding a constant

Property	Cosine	Correlation	Euclidean Distance
Invariant to scaling (multiplication)	Yes	Yes	No
Invariant to translation (addition)	No	Yes	No

EXAMPLE:

- $\mathbf{x} = (1, 2, 4, 3, 0, 0, 0)$,
- $\mathbf{y} = (1, 2, 3, 4, 0, 0, 0)$
- $\mathbf{y}_s = \mathbf{y} \cdot 2$ (scaled version of $\mathbf{y}$),
- $\mathbf{y}_t = \mathbf{y} + 5$ (translated version of $\mathbf{y}$)

Measure	$(\mathbf{x}, \mathbf{y})$	$(\mathbf{x}, \mathbf{y}_s)$	$(\mathbf{x}, \mathbf{y}_t)$
Cosine	0.9667	0.9667	0.7940
Correlation	0.9429	0.9429	0.9429
Euclidean Distance	1.4142	5.8310	14.2127

INFORMATION THEORY is a well-developed and fundamental discipline with broad applications.

- Some similarity measures are based on information theory:
 - **MUTUAL INFORMATION** in various versions
 - **MAXIMAL INFORMATION COEFFICIENT** (MIC) and related measures
 - general and can handle non-linear relationships
 - can be complicated and time intensive to compute

- Information relates to possible outcomes of an event
 - transmission of a message, flip of a coin, or measurement of a piece of data

- The more certain an outcome, the less information that it contains and vice-versa
 - for example, if a coin has two heads, then an outcome of heads provides no information
 - more quantitatively, the information is related the probability of an outcome
 - The smaller the probability of an outcome, the more information it provides and vice-versa
 - **ENTROPY** is the commonly used measure

ENTROPY, for

- a variable (event), X ,
- with n possible values (outcomes) $x_1, x_2, \dots, x_n$
- each outcome having probability $p_1, p_2, \dots, p_n$
- the entropy of X , $H(X)$, is given by

$$H(X) = - \sum_{i=1}^n p_i \log_2 p_i$$

- **ENTROPY** is between 0 and $\log_2 n$ and is measured in bits
 - thus, entropy is a measure of how many bits it takes to represent an observation of X on average

Given a coin $X \in \{head, tail\}$



$p = \text{probability of } X = \text{head}$

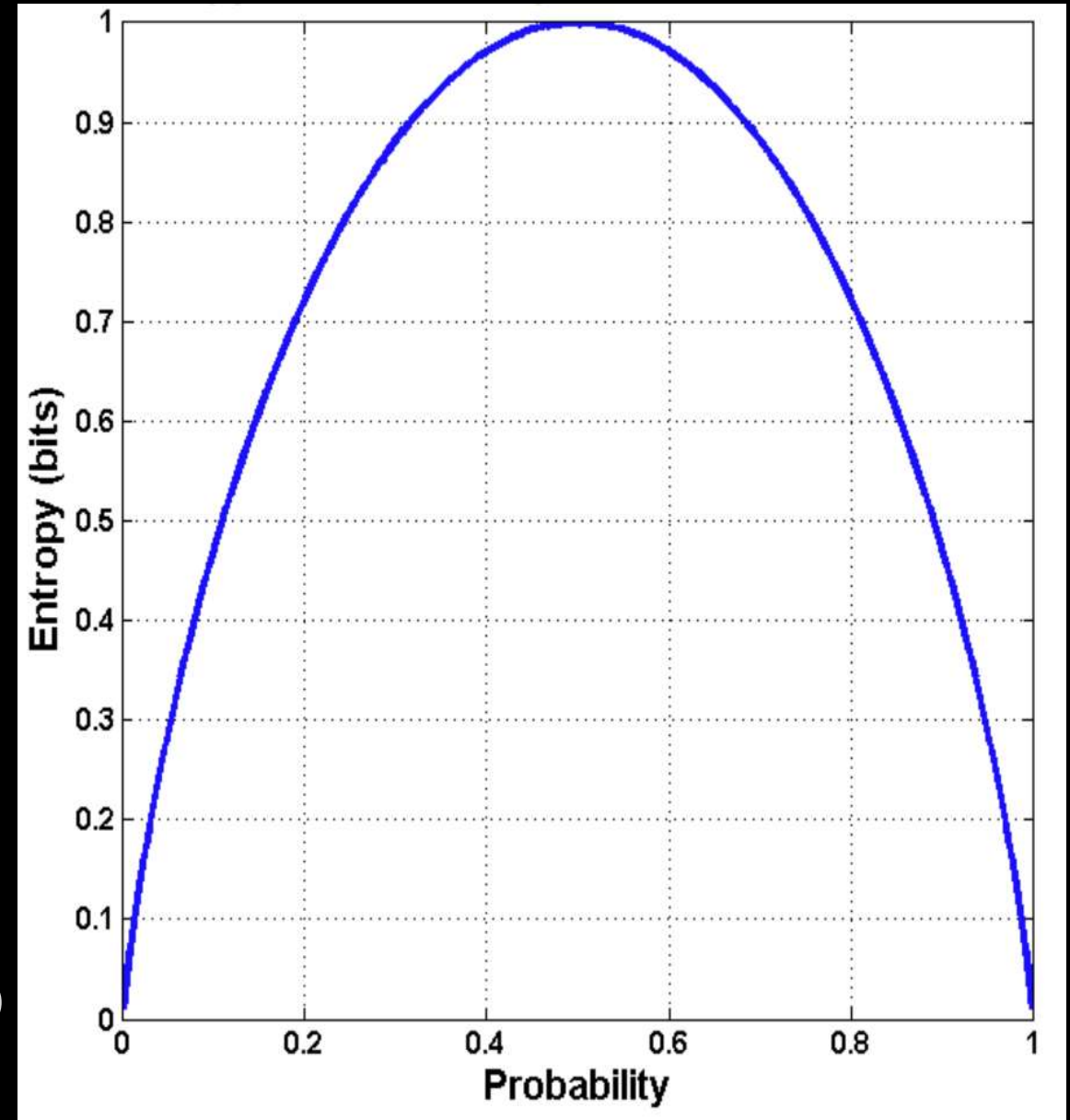
$1 - p = \text{probability of } X = \text{tail}$

$$H(X) = - \sum_{j=1}^m P(X = x_j) \log_2 P(X = x_j)$$

$$\begin{aligned} H(X) &= - \sum_{j=1}^2 P(X = x_j) \log_2 P(X = x_j) \\ &= -p \log_2 p - (1 - p) \log_2 (1 - p) \end{aligned}$$

$p = 0.5$ (fair coin) $\Rightarrow H(X) = 1$ (max uncertainty)

$p = 1.0$ $\Rightarrow H(X) = 0$ (min uncertainty)



Suppose we have

- a number of observations (m) of some attribute, X , e.g., the hair color of students in the class,
- where there are n different possible values
- and the number of observation in the i^{th} category is m_i
- then, for this sample

$$H(X) = - \sum_{i=1}^n \frac{m_i}{m} \log_2 \frac{m_i}{m}$$

Hair Color	Count	p	$-p \log_2 p$
Black	75	0.75	0.3113
Brown	15	0.15	0.4105
Blond	5	0.05	0.2161
Red	0	0.00	0
Other	5	0.05	0.2161
Total	100	1.0	1.1540

Hair Color $X \in \{Black, Brown, Blond, Red, Other\}$

$$H(X) = - \sum_{j=1}^5 P(X = x_j) \log_2 P(X = x_j) = 1.154$$

$$\max H(X) = \log_2 5 = 2.3219$$

For continuous attributes the computation is more difficult, i.e., we could apply discretization.

MUTUAL INFORMATION is used to measure the similarity between two sets of paired values.

- sometimes used in place of correlation, particularly when a nonlinear relationship is suspected between the pair of values
- a measure of how much information one set of values provides about another, given that the values come in pairs, e.g., height and weight
- if the **two sets of values are independent**, i.e., the value of one tells us nothing about the other, then their **mutual information is equal to 0**. On the other hand, if the **two sets of values are completely dependent**, i.e., knowing the value of one tells us the value of the other and vice-versa, then they have **maximum mutual information**
- mutual information does not have a maximum value, but we will define a normalized version of it that ranges between 0 and 1

MUTUAL INFORMATION

$$I(X, Y)$$

$$X \in \{x_1, \dots, x_m\} \quad Y \in \{y_1, \dots, y_n\}$$

$$H(X) = - \sum_{j=1}^m P(X = x_j) \log_2 P(X = x_j)$$

ENTROPY OF X

$$H(X) \in [0, \log_2 m]$$

$$H(Y) = - \sum_{k=1}^n P(Y = y_k) \log_2 P(Y = y_k)$$

ENTROPY OF Y

$$H(Y) \in [0, \log_2 n]$$

$$H(X, Y) = - \sum_{j=1}^m \sum_{k=1}^n P(X = x_j, Y = y_k) \log_2 P(X = x_j, Y = y_k)$$

**JOINT ENTROPY
OF X and Y**

$$H(X, Y) \in [0, \log_2 mn]$$

$$I(X, Y) = H(X) + H(Y) - H(X, Y)$$

MUTUAL INFORMATION

$$\max I(X, Y) = \log_2(\min(m, n))$$

$$H(X, Y) = H(Y, X) \Rightarrow I(X, Y) = I(Y, X)$$

(*) $0 \log_2 0 = 0$, by convention.

IS SYMMETRIC

Student Status	Count	p	$-p\log_2 p$
Undergrad	45	0.45	0.5184
Grad	55	0.55	0.4744
Total	100	1.00	0.9928

X

$$H(X) = -0.45 \log_2 0.45 - 0.55 \log_2 0.55 = 0.9928$$

Grade	Count	p	$-p\log_2 p$
A	35	0.35	0.5301
B	50	0.50	0.5000
C	15	0.15	0.4105
Total	100	1.00	1.4406

Y

$$H(Y) = -0.35 \log_2 0.35 - 0.50 \log_2 0.50 - 0.15 \log_2 0.15 = 1.4406$$

Student Status	Grade	Count	p	$-p\log_2 p$
Undergrad	A	5	0.05	0.2161
Undergrad	B	30	0.30	0.5211
Undergrad	C	10	0.10	0.3322
Grad	A	30	0.30	0.5211
Grad	B	20	0.20	0.4644
Grad	C	5	0.05	0.2161
Total		100	1.00	2.2710

X, Y

$$H(X, Y) = -0.05 \log_2 0.05 - 0.3 \log_2 0.3 - 0.1 \log_2 0.1 - 0.3 \log_2 0.3 - 0.2 \log_2 0.2 - 0.05 \log_2 0.05 = 2.2710$$

$$I(X, Y) = 0.9928 + 1.4406 - 2.2710 = 0.1624$$

$$\max I(X, Y) = \log_2(\min(2, 3)) = 1$$

$$\text{normalized } I(X, Y) = \frac{0.1624}{1} = 0.1624$$

(*) in this case there is no change

MUTUAL INFORMATION for nonlinear relationship

$$X = (-3, -2, -1, 0, 1, 2, 3)$$

$$Y = (9, 4, 1, 0, 1, 4, 9)$$

$$H(X) = -7 \left(\frac{1}{7} \log_2 \frac{1}{7} \right) = 2.8071$$

$$H(Y) = -3 \left(\frac{2}{7} \log_2 \frac{2}{7} \right) - \left(\frac{1}{7} \log_2 \frac{1}{7} \right) = 1.9502$$

$$H(X, Y) = -7 \left(\frac{1}{7} \log_2 \frac{1}{7} \right) = 2.8071$$

$$I(X, Y) = 2.8071 + 1.9502 - 2.8071 = 1.9502$$

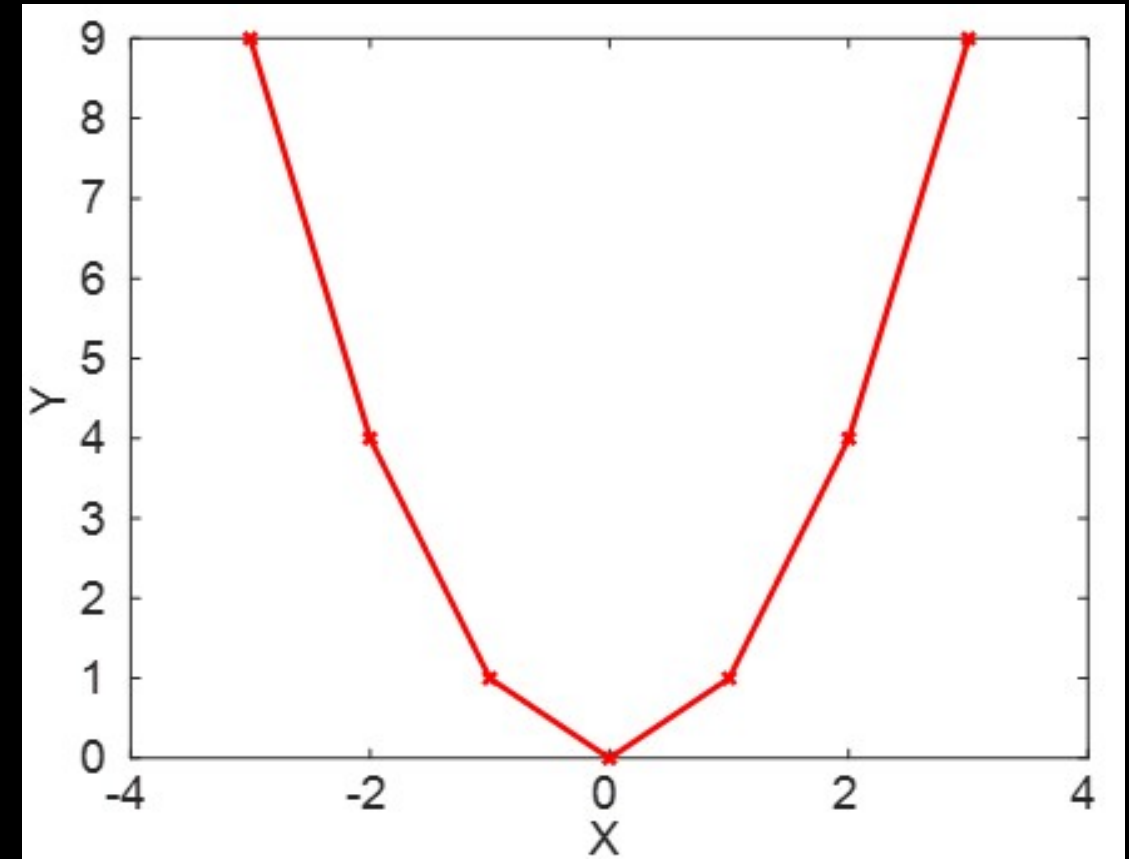
$$\text{normalized } I(X, Y) = \frac{1.9502}{2} = 0.9751$$

(*) X and Y are
strongly related

$$\max I(X, Y) = \log_2(\min(7, 4)) = 2$$

$$\text{Corr}(X, Y) = 0$$

(**) X and Y are unrelated



Sometimes attributes are of many different types, but an overall similarity is needed.

1: for the k^{th} attribute, compute a similarity, $s_k(\mathbf{x}, \mathbf{y})$, in the range $[0,1]$

2: define an indicator variable, δ_k , for the k^{th} attribute as follows:

$\delta_k = 0$ if the k^{th} attribute is an asymmetric attribute and both objects have a value of 0, or if one of the objects has a missing value for the k^{th} attribute

$\delta_k = 1$ otherwise

3. compute

$$\text{similarity}(\mathbf{x}, \mathbf{y}) = \frac{\sum_{k=1}^n \delta_k s_k(\mathbf{x}, \mathbf{y})}{\sum_{k=1}^n \delta_k}$$

May not want to treat all attributes the same.

- use non-negative weights w_k to compute

$$\text{similarity}(\mathbf{x}, \mathbf{y}) = \frac{\sum_{k=1}^n w_k \delta_k s_k(\mathbf{x}, \mathbf{y})}{\sum_{k=1}^n w_k \delta_k}$$

Can also define a weighted form of distance

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{k=1}^n w_k |x_k - y_k|^r \right)^{1/r}$$

Choice of the **RIGHT PROXIMITY MEASURE** depends on the domain

- What is the correct choice of proximity measure for the following situations?
 - Comparing documents using the frequencies of words
 - documents are considered similar if the word frequencies are similar
 - Comparing the temperature in Celsius of two locations
 - two locations are considered similar if the temperatures are similar in magnitude
 - Comparing two time series of temperature measured in Celsius
 - two time series are considered similar if their “shape” is similar, i.e., they vary in the same way over time, achieving minimums and maximums at similar times, etc...

- Domain of Application
 - similarity measures tend to be specific to the type of attribute and data
 - record data, images, graphs, sequences, 3D-protein structure, etc. tend to have different measures
- However, one can talk about various properties that you would like a proximity measure to have:
 - symmetry is a common one
 - tolerance to noise and outliers is another
 - ability to find more types of patterns?
 - many others possible
- The measure must be applicable to the data and produce results that agree with domain knowledge

RECAP

- **SIMILARITY – DISSIMILARITY**
- **EUCLIDEAN – MINKOWSKI – CITY BLOCK – MAHALANOBIS**
- **SIMPLE MATCHING COEFFICIENT – JACCARD – COSINE SIMILARITY**
- **CORRELATION – ENTROPY – MUTUAL INFORMATION**
- **SELECTING THE RIGHT PROXIMITY MEASURE**